Elastic Net

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Prerequisites: Linear Regression, Lasso, Ridge.

We define and process $\mathbf{X}, \mathbf{y}, \mathbf{w}$ similar to Linear Regression.

Elastic Net extends Linear Regression, Lasso, Ridge by combining both L_1 and L_2 regularization terms in the loss function. The resulting optimization problem is

$$\operatorname*{argmin}_{\mathbf{w}} \left(|\mathbf{y} - \mathbf{X}\mathbf{w}|^2 + \lambda_1 |\mathbf{w}|_1 + \lambda_2 |\mathbf{w}|_2^2 \right),$$

where

$$\sum_{j=1}^{p} |w_j|$$

is the L_1 norm of the coefficient vector, and

$$\sum_{j=1}^{p} w_j^2$$

is the squared L_2 norm. The parameters $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ control the strengths of the L_1 and L_2 penalties, respectively.

For each coefficient w_j, Elastic Net updates the coefficient using a soft-thresholding step:

$$w_j = \frac{S\left(\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}_{-j}\mathbf{w}_{-j}), \lambda_1\right)}{\mathbf{x}_j^\top \mathbf{x}_j + \lambda_2},$$

where

$$S(z, \lambda_1) = \begin{cases} z - \lambda_1, & z > \lambda_1, \\ 0, & |z| \leq \lambda_1, \\ z + \lambda_1, & z < -\lambda_1. \end{cases}$$

Here, $\mathbf{x}_j$ is the j-th feature column of $\mathbf{X}$, and $\mathbf{X}_{-j}\mathbf{w}_{-j}$ means the prediction using all coefficients except w_j.